

# Clase de Métodos Numéricos

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# Capítulo 1

## Inversa de una matriz

```
A=matrix(c(1,3,-2,1,1,4,-1,2,0,1,-1,4,2,6,1,2),nrow = 4,byrow = TRUE)
(A)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    3   -2    1
## [2,]    1    4   -1    2
## [3,]    0    1   -1    4
## [4,]    2    6    1    2
```

```
Aorig<- A
```

1)

```
I = diag(4);(I)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    0    0    0
## [2,]    0    1    0    0
## [3,]    0    0    1    0
## [4,]    0    0    0    1
```

2)

```
AInv <- cbind(A,I); (AInv)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    3   -2    1    1    0    0    0
## [2,]    1    4   -1    2    0    1    0    0
## [3,]    0    1   -1    4    0    0    1    0
## [4,]    2    6    1    2    0    0    0    1
```

```
A <- AInv
```

3)

```
l21 <- A[2,1]/A[1,1];(l21)
```

```
## [1] 1
```

```
131 <- A[3,1]/A[1,1];(131)
```

```
## [1] 0
```

```
141 <- A[4,1]/A[1,1];(141)
```

```
## [1] 2
```

4)

```
A[2,] <- A[2,]-121*A[1,];
```

```
A[3,] <- A[3,]-131*A[1,];
```

```
A[4,] <- A[4,]-141*A[1,];
```

```
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    3   -2    1    1    0    0    0
## [2,]    0    1    1    1   -1    1    0    0
## [3,]    0    1   -1    4    0    0    1    0
## [4,]    0    0    5    0   -2    0    0    1
```

5)

```
Atmp <- A[2,]
```

```
A[2,] <- A[3,]
```

```
A[3,] <- Atmp
```

```
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    3   -2    1    1    0    0    0
## [2,]    0    1   -1    4    0    0    1    0
## [3,]    0    1    1    1   -1    1    0    0
## [4,]    0    0    5    0   -2    0    0    1
```

6)

```
132 <- A[3,2]/A[2,2];(132)
```

```
## [1] 1
```

```
142 <- A[4,2]/A[2,2];(142)
```

```
## [1] 0
```

7)

```
A[3,] <- A[3,]-132*A[2,];
```

```
A[4,] <- A[4,]-142*A[2,];
```

```
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    3   -2    1    1    0    0    0
## [2,]    0    1   -1    4    0    0    1    0
## [3,]    0    0    2   -3   -1    1   -1    0
## [4,]    0    0    5    0   -2    0    0    1
```

8)

```
esc <- 1/A[3,3]
A[3,] <- esc*A[3,]
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    3   -2  1.0  1.0  0.0  0.0    0
## [2,]    0    1   -1  4.0  0.0  0.0  1.0    0
## [3,]    0    0    1 -1.5 -0.5  0.5 -0.5    0
## [4,]    0    0    5  0.0 -2.0  0.0  0.0    1
```

9)

```
143 <- A[4,3]/A[3,3]; (143)
```

```
## [1] 5
```

10)

```
A[4,] <- A[4,]-143*A[3,];
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    3   -2  1.0  1.0  0.0  0.0    0
## [2,]    0    1   -1  4.0  0.0  0.0  1.0    0
## [3,]    0    0    1 -1.5 -0.5  0.5 -0.5    0
## [4,]    0    0    0  7.5  0.5 -2.5  2.5    1
```

11)

```
esc <- 1/A[4,4]
A[4,] <- esc*A[4,]
(A)
```

```
##      [,1] [,2] [,3] [,4]      [,5]      [,6]      [,7]      [,8]
## [1,]    1    3   -2  1.0  1.00000000  0.00000000  0.00000000  0.00000000
## [2,]    0    1   -1  4.0  0.00000000  0.00000000  1.00000000  0.00000000
## [3,]    0    0    1 -1.5 -0.50000000  0.50000000 -0.50000000  0.00000000
## [4,]    0    0    0  1.0  0.06666667 -0.33333333  0.33333333  0.13333333
```

12)

```
134 <- A[3,4]/A[4,4]; (134); A[3,] <- A[3,]-134*A[4,]
```

```
## [1] -1.5
```

```
124 <- A[2,4]/A[4,4]; (124); A[2,] <- A[2,]-124*A[4,]
```

```
## [1] 4
```

```
114 <- A[1,4]/A[4,4]; (114); A[1,] <- A[1,]-114*A[4,]
```

```
## [1] 1
```

```
(A)
```

```
##      [,1] [,2] [,3] [,4]      [,5]      [,6]      [,7]      [,8]
## [1,]    1    3   -2    0  0.93333333  0.33333333 -0.33333333 -0.13333333
## [2,]    0    1   -1    0 -0.26666667  1.33333333 -0.33333333 -0.53333333
## [3,]    0    0    1    0 -0.40000000  0.00000000  0.00000000  0.20000000
```

```
## [4,] 0 0 0 1 0.06666667 -0.3333333 0.3333333 0.1333333
```

13)

```
123 <- A[2,3]/A[3,3]; (123); A[2,] <- A[2,]-123*A[3,]
```

```
## [1] -1
```

```
113 <- A[1,3]/A[3,3]; (113); A[1,] <- A[1,]-113*A[3,]
```

```
## [1] -2
```

(A)

```
##      [,1] [,2] [,3] [,4]      [,5]      [,6]      [,7]      [,8]
## [1,] 1 3 0 0 0.13333333 0.33333333 -0.33333333 0.26666667
## [2,] 0 1 0 0 -0.66666667 1.33333333 -0.33333333 -0.33333333
## [3,] 0 0 1 0 -0.40000000 0.00000000 0.00000000 0.20000000
## [4,] 0 0 0 1 0.06666667 -0.33333333 0.33333333 0.13333333
```

14)

```
112 <- A[1,2]/A[2,2]; (112); A[1,] <- A[1,]-112*A[2,]
```

```
## [1] 3
```

(A)

```
##      [,1] [,2] [,3] [,4]      [,5]      [,6]      [,7]      [,8]
## [1,] 1 0 0 0 2.13333333 -3.66666667 0.66666667 1.26666667
## [2,] 0 1 0 0 -0.66666667 1.33333333 -0.33333333 -0.33333333
## [3,] 0 0 1 0 -0.40000000 0.00000000 0.00000000 0.20000000
## [4,] 0 0 0 1 0.06666667 -0.33333333 0.33333333 0.13333333
```

15)

```
AInv <- A[,5:8]; (AInv)
```

```
##      [,1]      [,2]      [,3]      [,4]
## [1,] 2.13333333 -3.66666667 0.66666667 1.26666667
## [2,] -0.66666667 1.33333333 -0.33333333 -0.33333333
## [3,] -0.40000000 0.00000000 0.00000000 0.20000000
## [4,] 0.06666667 -0.33333333 0.33333333 0.13333333
```

16)

```
IdCalc <- Aorig%*%AInv; (IdCalc)
```

```
##      [,1]      [,2]      [,3]      [,4]
## [1,] 1.0000000e+00 -5.551115e-17 5.551115e-17 -2.775558e-17
## [2,] -3.053113e-16 1.000000e+00 1.110223e-16 0.000000e+00
## [3,] -5.551115e-17 0.000000e+00 1.000000e+00 0.000000e+00
## [4,] -5.273559e-16 -1.110223e-16 1.110223e-16 1.000000e+00
```



## Capítulo 2

# Factorización LU

### 2.1. Ejemplo 1

```
A=matrix(c(1,1,1,1,2,3,1,5,-1,1,-5,3,3,1,7,-2),byrow = TRUE,nrow = 4); (A)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    1    1    1
## [2,]    2    3    1    5
## [3,]   -1    1   -5    3
## [4,]    3    1    7   -2
```

```
b=matrix(c(10,31,-2,18),nrow = 4); (b)
```

```
##      [,1]
## [1,]   10
## [2,]   31
## [3,]   -2
## [4,]   18
```

```
Ab <- cbind(A,b); (Ab)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    1    1    1   10
## [2,]    2    3    1    5   31
## [3,]   -1    1   -5    3   -2
## [4,]    3    1    7   -2   18
```

Los pivotes se definen como  $a_{kk}^{(k)}$ , luego construimos los multiplicadores:  $l_{i,k} = a_{ik}^{(k)} / a_{kk}^{(k)}$ ,  $l_{21} = a_{21} / a_{11}$  y  $l_{31} = a_{31} / a_{11}$  y  $l_{41} = a_{41} / a_{11}$

```
A <- Ab;
```

```
l21 <- A[2,1]/A[1,1]; (l21)
```

```
## [1] 2
```

```
l31 <- A[3,1]/A[1,1]; (l31)
```

```
## [1] -1
```

```
141 <- A[4,1]/A[1,1];(141)
```

```
## [1] 3
```

Construimos la matriz  $U$ , donde las entradas  $a_{ij}^{(k+1)} = a_{ik}^{(k)} - l_{ik} \times a_{kj}^{(k)}$

```
A[2,] <- A[2,]-121*A[1,]; (A[2,])
```

```
## [1] 0 1 -1 3 11
```

```
A[3,] <- A[3,]-131*A[1,]; (A[3,])
```

```
## [1] 0 2 -4 4 8
```

```
A[4,] <- A[4,]-141*A[1,]; (A[4,])
```

```
## [1] 0 -2 4 -5 -12
```

Es decir la matriz resultante es:

```
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    1    1    1   10
## [2,]    0    1   -1    3   11
## [3,]    0    2   -4    4    8
## [4,]    0   -2    4   -5   -12
```

entonces el pivote es  $A_{22} = 1$ , calculemos los  $l_{32}$  y  $l_{42}$

```
132 <- A[3,2]/A[2,2];(132)
```

```
## [1] 2
```

```
142 <- A[4,2]/A[2,2];(142)
```

```
## [1] -2
```

Haciendo cero debajo del pivote

```
A[3,] <- A[3,]-132*A[2,]; (A[3,])
```

```
## [1] 0 0 -2 -2 -14
```

```
A[4,] <- A[4,]-142*A[2,]; (A[4,])
```

```
## [1] 0 0 2 1 10
```

La matriz  $A$  queda de la forma:

```
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    1    1    1   10
## [2,]    0    1   -1    3   11
## [3,]    0    0   -2   -2  -14
## [4,]    0    0    2    1   10
```

por tanto el pivote es  $A_{33} = -2$ , y resta calcular  $l_{43}$

```
143 <- A[4,3]/A[3,3];(143)
```

```
## [1] -1
```

haciendo cero debajo del pivote

```
A[4,] <- A[4,]-143*A[3,]; (A[4,])
```

```
## [1] 0 0 0 -1 -4
```

Por lo tanto la matriz  $A$  resultante es

```
(A)
```

```
##      [,1] [,2] [,3] [,4] [,5]
## [1,]    1    1    1    1   10
## [2,]    0    1   -1    3   11
## [3,]    0    0   -2   -2  -14
## [4,]    0    0    0   -1   -4
```

entonces podemos construir la matriz  $L$  con los valores  $l_{ij}$  calculados en los pasos anteriores

```
L=diag(4); (L)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    0    0    0
## [2,]    0    1    0    0
## [3,]    0    0    1    0
## [4,]    0    0    0    1
```

```
L[2,1] <- 121
L[3,1] <- 131
L[4,1] <- 141
L[3,2] <- 132
L[4,2] <- 142
L[4,3] <- 143
(L)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    0    0    0
## [2,]    2    1    0    0
## [3,]   -1    2    1    0
## [4,]    3   -2   -1    1
```

```
U <- A[,1:4]; (U)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    1    1    1
## [2,]    0    1   -1    3
## [3,]    0    0   -2   -2
## [4,]    0    0    0   -1
```

Verifiquemos que efectivamente  $A = LU$

```
Acalculada <- L%*%U; (Acalculada)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    1    1    1
## [2,]    2    3    1    5
## [3,]   -1    1   -5    3
## [4,]    3    1    7   -2
```

## 2.2. Ejemplo 2

Apliquemos lo desarrollado para la siguiente matriz  $A$

```
A=matrix(c(4,0,1,1,3,1,3,1,0,1,2,0,3,2,4,1),nrow = 4,byrow = TRUE); (A)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    4    0    1    1
## [2,]    3    1    3    1
## [3,]    0    1    2    0
## [4,]    3    2    4    1
```

```
b=matrix(c(5,6,13,1),nrow = 4);(b)
```

```
##      [,1]
## [1,]    5
## [2,]    6
## [3,]   13
## [4,]    1
```

1)

```
Ab <- cbind(A,b)
A <- Ab;
```

2)

```
l21 <- A[2,1]/A[1,1];(l21)
```

```
## [1] 0.75
```

```
l31 <- A[3,1]/A[1,1];(l31)
```

```
## [1] 0
```

```
l41 <- A[4,1]/A[1,1];(l41)
```

```
## [1] 0.75
```

3)

```
A[2,] <- A[2,]-l21*A[1,]; (A[2,])
```

```
## [1] 0.00 1.00 2.25 0.25 2.25
```

```
A[3,] <- A[3,]-l31*A[1,]; (A[3,])
```

```
## [1] 0 1 2 0 13
```

```
A[4,] <- A[4,]-l41*A[1,]; (A[4,])
```

```
## [1] 0.00 2.00 3.25 0.25 -2.75
```

4)

```
l32 <- A[3,2]/A[2,2];(l32)
```

```
## [1] 1
```

```
l42 <- A[4,2]/A[2,2];(l42)
```

```
## [1] 2
```

5)

```
A[3,] <- A[3,]-132*A[2,];(A[3,])
```

```
## [1] 0.00 0.00 -0.25 -0.25 10.75
```

```
A[4,] <- A[4,]-142*A[2,];(A[4,])
```

```
## [1] 0.00 0.00 -1.25 -0.25 -7.25
```

6)

```
143 <- A[4,3]/A[3,3];(143)
```

```
## [1] 5
```

7)

```
A[4,] <- A[4,]-143*A[3,];(A[4,])
```

```
## [1] 0 0 0 1 -61
```

8)

```
L=diag(4);(L)
```

```
##      [,1] [,2] [,3] [,4]
## [1,] 1    0    0    0
## [2,] 0    1    0    0
## [3,] 0    0    1    0
## [4,] 0    0    0    1
```

9)

```
L[2,1] <- 121
L[3,1] <- 131
L[4,1] <- 141
L[3,2] <- 132
L[4,2] <- 142
L[4,3] <- 143;
(L)
```

```
##      [,1] [,2] [,3] [,4]
## [1,] 1.00 0    0    0
## [2,] 0.75 1    0    0
## [3,] 0.00 1    1    0
## [4,] 0.75 2    5    1
```

10)

```
U <- A[,1:4];(U)
```

```
##      [,1] [,2] [,3] [,4]
## [1,] 4    0 1.00 1.00
## [2,] 0    1 2.25 0.25
## [3,] 0    0 -0.25 -0.25
## [4,] 0    0 0.00 1.00
```

11)

```
Acalculada <- L%*%U; (Acalculada)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    4    0    1    1
## [2,]    3    1    3    1
## [3,]    0    1    2    0
## [4,]    3    2    4    1
```

## 2.3. Ejercicio

Aplicar la factorización  $LU$  a la matriz  $A$  definida por

```
A=matrix(c(2,3,2,4,
           4,10,-4,0,
           -3,-2,-5,-2,
           -2,4,4,-7),nrow = 4,byrow = TRUE)
```

(A)

```
##      [,1] [,2] [,3] [,4]
## [1,]    2    3    2    4
## [2,]    4   10   -4    0
## [3,]   -3   -2   -5   -2
## [4,]   -2    4    4   -7
```

## Capítulo 3

# Notas importantes

**Nota 1** Si se fija  $l_{ii} = 1$  se le denomina factorización *Doolittle*.

**Nota 2** Una matriz  $A$  tiene factorización de Doolittle si y sólo si se le puede aplicar el método de eliminación de Gauss sin pivoteo.

**Nota 3** Si  $U = L^T$  entonces la factorización se le denomina *factorización de Cholesky*.

**Nota 4** Si la matriz es simétrica y definida positiva, entonces  $A = LL^T$ .

